V COVERINGS ARE UNIVERSALLY SUBMERSIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [4045, 4057] and proof(s) [[4059, 4081]]; this spec covers the complete environment instead (statement 4045-4057, proof span(s) [(4059, 4081)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The argument cites the in-chapter universally-submersive characterization and applies it; the citation is disclosed in the proof text.

Source reference. Original reference 040H: sources/stacks-chow/tags.txt, source lines 5256–5256. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0ETJ: sources/stacks-chow/tags.txt, source lines 18519–18519. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0ETN: sources/stacks-chow/tags.txt, source lines 18523–18523. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00HY: sources/stacks-chow/tags.txt, source lines 660–660. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\{f_i : X_i \to X\}_{i \in I}$ be a V covering. Then*

$$\coprod_{i \in I} f_i : \coprod_{i \in I} X_i \longrightarrow X$$

is a universally submersive morphism of schemes (Morphisms, Definition 040H).

Proof. We will use without further mention that the base change of a V covering is a V covering (Lemma 0ETJ). In particular it suffices to show that the morphism is submersive. Being submersive is clearly Zariski local on the base. Thus we may assume X is affine. Then $\{X_i \to X\}$ can be refined by a standard V covering $\{Y_j \to X\}$. If we can show that $\coprod Y_j \to X$ is submersive, then since there is a factorization $\coprod Y_j \to \coprod X_i \to X$ we conclude that $\coprod X_i \to X$ is submersive. Set $Y = \coprod Y_j$ and consider the morphism of affines $f : Y \to X$. By Lemma 0ETN we know that we can lift any specialization $x' \rightsquigarrow x$ in X to some specialization $y' \rightsquigarrow y$ in Y. Thus if $T \subset X$ is a subset such that $f^{-1}(T)$ is closed in Y, then $T \subset X$ is closed under specialization. Since $f^{-1}(T) \subset Y$ with the reduced induced closed subscheme structure is an affine scheme, we conclude that $T \subset X$ is closed by Algebra, Lemma 00HY. Hence f is submersive. □

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